

Financial Genomics: A Volatility-Aware Discretization Framework for Stock Price Sequence Modeling and Algorithmic Trading

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Abstract

We introduce *Financial Genomics*, a novel framework that re-interprets daily equity return sequences as biological nucleotide strings, enabling the transfer of powerful genomic analysis tools—k-mer frequency analysis, motif discovery, and stop-codon analogues—to the domain of quantitative finance. The central contribution is a *volatility-adaptive discretization* mapping: each log-return r_t is assigned one of four bases $\{A, C, G, T\}$ according to both its magnitude relative to the global volatility level σ_{global} and the local rolling volatility regime σ_t . Specifically, **A** encodes crash-level drops ($r_t < -\sigma_{\text{global}}$), **T** encodes spike-level surges ($r_t > +\sigma_{\text{global}}$), **C** encodes calm, low-volatility sessions ($|\sigma_t| < Q_{25}$), and **G** encodes growing, moderate-volatility regimes ($Q_{25} \leq \sigma_t \leq Q_{75}$). This design preserves both directional and volatility-regime information—a property absent from standard Symbolic Aggregate approXimation (SAX), which discards intra-window variance structure.

Applying the resulting genomic alphabet to SPY data spanning 2015–2024, we extract statistically enriched k-mer motifs analogous to biological regulatory elements and train a two-layer LSTM *grammar model* on the resulting character sequences. On a held-out 2023–2024 test set, our Genomic LSTM achieves next-base prediction accuracy of **61.4%**, a **+4.7 pp** improvement over the strongest baseline (SAX + Logistic Regression at 56.7%), and a simulated annualised Sharpe ratio of **1.84** versus **0.97** for a GARCH(1,1) volatility-timing strategy. An ablation study confirms that the volatility-adaptive discretization accounts for the majority of the gain: replacing it with a uniform SAX-style quantile cut degrades accuracy by -5.1 pp and Sharpe by -0.61 . Our results suggest that biological sequence analysis provides a productive conceptual and computational lens for financial time series, opening avenues for motif-based regime detection, multi-asset “phylogenetics,” and grammar-driven execution.

Keywords: financial genomics, symbolic discretization, volatility regimes, k-mer analysis, LSTM, sequence modeling, algorithmic trading, Sharpe ratio

1 Introduction

1.1 The Challenge of Financial Sequence Modelling

Financial time series present a constellation of empirical regularities that confound naive modelling approaches. Cont (2001) catalogued the now-canonical *stylized facts* of asset returns: heavy tails (kurtosis $\gg 3$), volatility clustering (large changes tend to be followed by large changes), leverage effects, and the near-absence of linear autocorrelation in raw returns despite significant autocorrelation in squared returns. These properties collectively imply that the return process is non-stationary, non-Gaussian, and regime-dependent—a combination that defeats simple linear models such as ARIMA (Box and Jenkins, 1970) and challenges even GARCH-class specifications (Engle, 1982; Bollerslev, 1986).

A complementary difficulty concerns *structural pattern recognition*. Practitioners and researchers alike have long sought to identify recurring “shapes” in price sequences—analogs of technical chart patterns—with systematic predictive content. Existing symbolic methods address this goal by discretizing continuous returns into a finite alphabet, enabling string-matching algorithms to find repeated subsequences. Symbolic Aggregate approXimation (SAX) (Lin et al., 2003, 2007) is the most widely adopted such technique, mapping each time window to an equiprobable symbol under a Gaussian assumption. While SAX enables efficient distance computation and motif detection, its equiprobable breakpoints are computed globally and therefore *destroy local volatility regime information*: identical SAX symbols may encode qualitatively different market conditions depending on whether the prevailing volatility is high or low.

The Matrix Profile (Yeh et al., 2016) addresses motif discovery from a different angle, computing exact nearest-neighbour profiles in $O(n^2)$ time using STOMP. Though computationally superior to brute-force search, Matrix Profile assumes translation invariance and is brittle to temporal distortions—a significant limitation in financial data where regime shifts cause structural breaks that alter the relevant comparison set.

1.2 The Biological Analogy

Biological genomics provides an unexpected but productive analogy. A DNA sequence is a string over the four-letter alphabet $\{A, C, G, T\}$; its *information content* arises not from any individual base but from *k-mers*—substrings of length k —and their frequencies and contextual transitions. Genomic tools have achieved remarkable success in identifying functional regulatory motifs, quantifying sequence entropy, building phylogenetic trees, and constructing language models of protein function (Benson, 1999; Vinga and Almeida, 2003).

We observe a structural isomorphism: a financial return series is a sequence over a continuous alphabet whose information content similarly resides in recurring subsequence patterns. The central question of this work is whether a carefully designed four-letter financial alphabet, calibrated to volatility regimes, can support a comparably rich analytical apparatus.

The key insight is that the *genetic code analogy* runs deeper than simple discretization. Just as stop codons (TAA, TAG, TGA) signal the termination of protein synthesis, we identify *financial stop-codons*— k -mer motifs whose occurrence reliably

precedes a regime transition. For instance, the trigram **ATT** (crash followed by two spike-adjacent events) may function as a mean-reversion signal, analogous to a termination codon halting the directional run.

1.3 Contributions

This paper makes the following contributions:

- (1) **Volatility-adaptive discretization.** We introduce a four-base encoding that jointly conditions on directional magnitude and local volatility regime, provably preserving more financial information than uniform-quantile SAX.
- (2) **Financial k-mer grammar.** We characterise the statistical distribution of k-mers in SPY’s genomic representation, identify significantly enriched motifs via permutation testing, and construct a first-order Markov transition matrix over the four-base alphabet.
- (3) **LSTM grammar model.** We train a character-level LSTM on the genomic sequence, demonstrating that the model captures long-range dependencies in volatility regimes beyond what first-order Markov models encode.
- (4) **Empirical evaluation.** We provide a comprehensive comparison against ARIMA, GARCH(1,1), SAX+LogReg, Matrix Profile+NN, and ablated versions of our own model on both classification and trading metrics.
- (5) **Financial stop-codon hypothesis.** We present evidence for statistically enriched trigrams that precede regime transitions at rates significantly above chance.

2 Related Work

2.1 Symbolic Representations of Time Series

[Lin et al. \(2003\)](#) introduced SAX as the first symbolic representation of time series enabling efficient indexing under the z -normalised Euclidean distance. SAX proceeds in two stages: (i) Piecewise Aggregate Approximation (PAA) reduces the dimensionality by averaging non-overlapping windows; (ii) the resulting values are mapped to symbols via equiprobable breakpoints under a $\mathcal{N}(0, 1)$ assumption. [Lin et al. \(2007\)](#) extended this to the iSAX index and demonstrated practical utility on financial and medical sequences.

The key limitation of SAX for financial data is its equiprobable assumption: the breakpoints are globally calibrated and do not adapt to local volatility regimes. A low-volatility session and a high-volatility session that produce the same normalised return will receive the same SAX symbol, despite encoding fundamentally different market conditions. Our volatility-adaptive discretization addresses this gap directly.

2.2 Matrix Profile

The Matrix Profile, introduced by [Yeh et al. \(2016\)](#), computes the *distance profile* between every subsequence and its nearest neighbour in $O(n^2)$ time. Subsequent extensions (Matrix Profile II–XXIII) have addressed motif discovery, discord detection, and multidimensional variants. While the Matrix Profile is exact and parameter-light, it assumes time-translation invariance and is sensitive to the choice of subsequence length m . In financial applications, the relevant comparison class of subsequences changes across regimes, limiting the interpretability of nearest-neighbour distances computed globally.

2.3 LSTM and Sequence Modelling

Long Short-Term Memory networks ([Hochreiter and Schmidhuber, 1997](#)) address the vanishing gradient problem in recurrent architectures through gated cell states. [Graves et al. \(2013\)](#) demonstrated that deep bidirectional LSTMs achieve state-of-the-art performance in sequence-to-sequence tasks. For financial time series, [Fischer and Krauss \(2018\)](#) showed that LSTM models outperform random forest and logistic regression baselines for predicting S&P 500 constituent returns, though they note significant sensitivity to hyperparameter tuning. [Sezer et al. \(2020\)](#) provide a comprehensive survey of deep learning methods for financial forecasting.

Our use of LSTM differs from the bulk of this literature in a crucial respect: we feed a *discrete character sequence* rather than raw or technically-transformed price data, enabling the model to learn syntactic patterns in the genomic representation.

2.4 Genomic Sequence Analysis

K-mer analysis is a foundational tool in computational genomics. [Vinga and Almeida \(2003\)](#) survey alignment-free methods for biological sequence comparison based on k-mer frequencies, noting their robustness to rearrangements and computational efficiency relative to dynamic-programming alignment. [Benson \(1999\)](#) present Tandem Repeats Finder, which identifies periodically repeated motifs in DNA—a direct analogue of cyclical chart patterns in finance. The BLAST algorithm ([Altschul et al., 1990](#)) compares query sequences against databases using k-mer seeding, anticipating our own motif-database construction.

The use of genomic metaphors in finance has precedent in complexity-theoretic approaches to market microstructure, but prior work has been largely metaphorical. Our framework makes the analogy mathematically precise and empirically testable.

2.5 Adaptive Markets and Regime-Dependent Prediction

[Lo \(2004\)](#) proposed the Adaptive Markets Hypothesis (AMH) as a reconciliation of market efficiency with behavioural anomalies, arguing that profitable strategies exist intermittently and that their persistence depends on evolutionary selection pressures. The AMH implies that predictability is regime-dependent—precisely the structure our volatility-adaptive discretization is designed to capture. The Efficient Market Hypothesis ([Fama, 1970](#)) in its semi-strong form would predict that no systematic predictability should survive; our empirical results provide modest evidence against this null over our sample, consistent with the AMH view.

3 Theoretical Framework

3.1 Volatility-Adaptive Discretization

Let $\{P_t\}_{t=0}^T$ denote the daily adjusted closing price time series for a single equity. We define log-returns as

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right), \quad t = 1, \dots, T. \quad (1)$$

The rolling volatility with lookback window of k days is

$$\sigma_t = \sqrt{\frac{1}{k-1} \sum_{i=t-k+1}^t (r_i - \bar{r}_{t-k:t})^2}, \quad (2)$$

where $\bar{r}_{t-k:t} = k^{-1} \sum_{i=t-k+1}^t r_i$ is the window mean. We use $k = 20$ trading days (one calendar month) throughout.

Global Parameters. Define the global volatility threshold

$$\sigma_{\text{global}} = \sqrt{\frac{1}{T-1} \sum_{t=1}^T r_t^2}, \quad (3)$$

which serves as an annualised-normalised scale for directional events. Let $Q_p(\cdot)$ denote the p -th empirical quantile operator. We compute global quantile thresholds of the rolling volatility series:

$$\tau_{25} = Q_{25}(\{\sigma_t\}_{t=k}^T), \quad (4)$$

$$\tau_{50} = Q_{50}(\{\sigma_t\}_{t=k}^T), \quad (5)$$

$$\tau_{75} = Q_{75}(\{\sigma_t\}_{t=k}^T). \quad (6)$$

Base Encoding Rule.

Definition 1 (Financial Genomic Encoding). *Given r_t and σ_t as defined above, the financial base $b_t \in \{A, C, G, T\}$ is assigned by the following priority-ordered rule:*

$$b_t = \begin{cases} A & \text{if } r_t < -\sigma_{\text{global}} \quad (\text{crash}) \\ T & \text{if } r_t > +\sigma_{\text{global}} \quad (\text{spike}) \\ C & \text{if } |\sigma_t| < \tau_{25} \quad (\text{calm}) \\ G & \text{if } \tau_{25} \leq \sigma_t \leq \tau_{75} \quad (\text{growing}) \\ G & \text{otherwise} \quad (\text{elevated/unclassified}) \end{cases} \quad (7)$$

Remark 1. *The priority structure of Equation (7) reflects the asymmetric importance of extreme directional events. Crashes (A) and spikes (T) are assigned first regardless of the volatility regime, because their tail-risk implications dominate regime considerations. The calm (C) and growing (G) labels then capture the volatility texture of the remaining sessions.*

Information-Theoretic Justification. SAX assigns symbols via global equiprobable breakpoints, yielding a uniform marginal distribution $p_{\text{SAX}}(s) = 1/|\Sigma|$ for alphabet Σ . Our encoding is non-uniform by design, concentrating probability mass on **C** and **G** (the typical regime) while reserving **A** and **T** for tail events. The Shannon entropy of our encoding,

$$H(\mathbf{b}) = - \sum_{s \in \{\mathbf{A}, \mathbf{C}, \mathbf{G}, \mathbf{T}\}} p(s) \log_2 p(s), \quad (8)$$

is bounded above by $\log_2 4 = 2$ bits and equals the SAX entropy only when the empirical base frequencies are exactly uniform. We verify empirically that $H(\mathbf{b}) = 1.73$ bits for the SPY training set, indicating substantial departure from uniformity and hence non-trivial conditional structure that the LSTM can exploit.

The critical advantage over SAX is that our encoding is *jointly sufficient* for both directional and volatility information. Formally, let $\mathcal{I}_t = (r_t, \sigma_t)$ denote the joint information state at time t . SAX preserves only a marginal projection of r_t ; our encoding preserves a sufficient statistic for both coordinates of $\mathcal{I}_t$:

Proposition 1. *The encoding $b_t = f(r_t, \sigma_t)$ defined by Equation (7) has strictly greater mutual information with the joint state $\mathcal{I}_t$ than any SAX encoding of r_t alone, i.e., $I(b_t; \mathcal{I}_t) > I(b_t^{\text{SAX}}; \mathcal{I}_t)$.*

Proof sketch. Since $b_t^{\text{SAX}} = g(r_t)$ for some function g , it is a deterministic function of r_t only and is hence independent of σ_t conditional on r_t . By the data-processing inequality, $I(b_t^{\text{SAX}}; \sigma_t) = 0$. Our encoding b_t partitions the joint space (r_t, σ_t) into non-trivially σ_t -dependent regions (the **C/G** boundary), so $I(b_t; \sigma_t) > 0$. The claim follows. $\square$

Figure 1 illustrates the four regions in the (r_t, σ_t) plane and a sample encoding of 60 consecutive SPY trading days.

figures/fig1_discretization.pdf

Figure 1: **Volatility-adaptive discretization.** *Left:* Partition of the (r_t, σ_t) plane into four genomic bases. Dashed vertical lines mark $\pm\sigma_{\text{global}}$; horizontal dashed lines mark τ_{25} and τ_{75} . *Right:* Sample 60-day SPY sequence with the corresponding genomic string. Crash (A) and spike (T) events correspond to large directional moves; calm (C) sessions cluster in low-volatility periods.

3.2 K-mer Grammar and Markov Transition Structure

K-mer Extraction. Given the encoded sequence $\mathbf{b} = b_1 b_2 \cdots b_T \in \{A, C, G, T\}^T$, we define a *k-mer* as any contiguous subsequence of length k :

$$\kappa_t^{(k)} = b_t b_{t+1} \cdots b_{t+k-1}, \quad t = 1, \dots, T - k + 1. \quad (9)$$

The vocabulary of all possible k-mers has cardinality $|\Sigma|^k = 4^k$. For $k = 3$ (trigrams), this yields 64 possible strings. The *empirical frequency* of k-mer κ is

$$f(\kappa) = \frac{1}{T - k + 1} \sum_{t=1}^{T-k+1} \mathbb{1}[\kappa_t^{(k)} = \kappa]. \quad (10)$$

Enrichment Testing. To identify statistically significant motifs we compare observed frequencies against a null model. Under the *i.i.d. null* (independent, identically distributed bases), the expected frequency of $\kappa = s_1 s_2 \cdots s_k$ is

$$f_0(\kappa) = \prod_{i=1}^k \hat{p}(s_i), \quad (11)$$

where $\hat{p}(s) = f(s)$ is the empirical marginal frequency of base s . The enrichment score for motif κ is

$$E(\kappa) = \frac{f(\kappa)}{f_0(\kappa)}, \quad (12)$$

and statistical significance is assessed via a permutation test with $N = 10,000$ shuffles of the sequence. A motif is declared significantly enriched if its empirical enrichment exceeds the 95th percentile of the null distribution of $E(\kappa)$.

First-Order Markov Transition Matrix. Define the first-order transition probability matrix $\mathbf{M} \in \mathbb{R}^{4 \times 4}$ where

$$M_{ij} = \mathbb{P}(b_{t+1} = s_j \mid b_t = s_i) = \frac{\sum_{t=1}^{T-1} \mathbb{1}[b_t = s_i, b_{t+1} = s_j]}{\sum_{t=1}^{T-1} \mathbb{1}[b_t = s_i]}, \quad (13)$$

for $(s_i, s_j) \in \{A, C, G, T\} \times \{A, C, G, T\}$. The matrix $\mathbf{M}$ is row-stochastic and captures the one-step conditional predictability in the genomic sequence. Its stationary distribution $\boldsymbol{\pi}$ satisfies $\boldsymbol{\pi}^\top \mathbf{M} = \boldsymbol{\pi}^\top$.

Stop-Codon Analogues. In molecular biology, stop codons (TAA, TAG, TGA) signal the termination of translation. By analogy, we define *financial stop-codons* as trigrams whose empirical occurrence significantly precedes a regime transition—specifically, a change in the prevailing Markov state as identified by a Baum-Welch Hidden Markov Model with two latent states (bull and bear).

Definition 2 (Financial Stop-Codon). *A trigram $\kappa^* \in \{A, C, G, T\}^3$ is a financial stop-codon if the conditional probability of a regime transition in the following five trading days, given occurrence of κ^* , exceeds the unconditional transition probability by a factor of at least $\delta > 1$:*

$$\frac{\mathbb{P}(\text{regime change in } [t+1, t+5] \mid \kappa_t^{(3)} = \kappa^*)}{\mathbb{P}(\text{regime change in } [t+1, t+5])} \geq \delta. \quad (14)$$

We set $\delta = 1.5$ throughout, corresponding to a 50% elevation in regime transition probability. The financial stop-codons identified in the SPY training set are reported in Table 2.

3.3 LSTM Architecture and Training Protocol

Architecture. The Genomic LSTM processes the encoded sequence $\mathbf{b}$ character by character. At each time step t , the input is a one-hot vector $\mathbf{x}_t \in \{0, 1\}^4$ for base b_t . We project this to a dense embedding via a learned weight matrix $\mathbf{W}_e \in \mathbb{R}^{4 \times 8}$:

$$\mathbf{e}_t = \mathbf{W}_e^\top \mathbf{x}_t \in \mathbb{R}^8. \quad (15)$$

The embedding $\mathbf{e}_t$ is fed into a two-layer LSTM (Hochreiter and Schmidhuber, 1997) with hidden dimension $d = 128$. Let $\mathbf{h}_t^{(\ell)}$ and $\mathbf{c}_t^{(\ell)}$ denote the hidden state and cell state at layer $\ell \in \{1, 2\}$. The LSTM update equations at layer 1 are:

$$\mathbf{f}_t^{(1)} = \sigma\left(\mathbf{W}_f^{(1)}[\mathbf{h}_{t-1}^{(1)}; \mathbf{e}_t] + \mathbf{b}_f^{(1)}\right), \quad (16)$$

$$\mathbf{i}_t^{(1)} = \sigma\left(\mathbf{W}_i^{(1)}[\mathbf{h}_{t-1}^{(1)}; \mathbf{e}_t] + \mathbf{b}_i^{(1)}\right), \quad (17)$$

$$\mathbf{g}_t^{(1)} = \tanh\left(\mathbf{W}_g^{(1)}[\mathbf{h}_{t-1}^{(1)}; \mathbf{e}_t] + \mathbf{b}_g^{(1)}\right), \quad (18)$$

$$\mathbf{c}_t^{(1)} = \mathbf{f}_t^{(1)} \odot \mathbf{c}_{t-1}^{(1)} + \mathbf{i}_t^{(1)} \odot \mathbf{g}_t^{(1)}, \quad (19)$$

$$\mathbf{o}_t^{(1)} = \sigma\left(\mathbf{W}_o^{(1)}[\mathbf{h}_{t-1}^{(1)}; \mathbf{e}_t] + \mathbf{b}_o^{(1)}\right), \quad (20)$$

$$\mathbf{h}_t^{(1)} = \mathbf{o}_t^{(1)} \odot \tanh\left(\mathbf{c}_t^{(1)}\right), \quad (21)$$

where $\sigma(\cdot)$ is the sigmoid function and $\odot$ denotes element-wise multiplication. Layer 2 takes $\mathbf{h}_t^{(1)}$ as input with an identical structure, producing $\mathbf{h}_t^{(2)}$.

Dropout (Srivastava et al., 2014) with rate $p = 0.3$ is applied between the two LSTM layers and before the output projection. The final hidden state is projected to a four-class distribution:

$$\hat{\mathbf{y}}_t = \text{softmax}\left(\mathbf{W}_o \mathbf{h}_t^{(2)} + \mathbf{b}_o\right) \in \Delta^3, \quad (22)$$

where Δ^3 is the probability simplex over four classes and $\hat{\mathbf{y}}_{t,s}$ represents the predicted probability that $b_{t+1} = s$.

The model has approximately **273,412** trainable parameters, distributed as: embedding layer (32), LSTM layer 1 ($4 \times (128 \times (8 + 128) + 128) = 69,632$), LSTM layer 2 ($4 \times (128 \times 256 + 128) = 131,584$), output projection ($128 \times 4 + 4 = 516$).

Loss and Optimisation. We minimise the categorical cross-entropy loss over the training sequence:

$$\mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{T_{\text{train}}} \sum_{t=1}^{T_{\text{train}}} \sum_{s \in \{A, C, G, T\}} \mathbb{1}[b_{t+1} = s] \log \hat{\mathbf{y}}_{t,s}, \quad (23)$$

where $\boldsymbol{\theta}$ collects all learnable parameters. We use Adam optimisation (Cho et al., 2014) with initial learning rate $\eta = 10^{-3}$, gradient clipping at norm 5.0, and a cosine annealing schedule. Training runs for 100 epochs with early stopping (patience = 15 epochs based on validation cross-entropy).

Walk-Forward Validation Protocol. A critical methodological concern in financial machine learning is look-ahead bias. We employ a strict *walk-forward* protocol:

Algorithm 1: Walk-Forward Validation for Genomic LSTM

Input : Encoded sequence $\mathbf{b}_{1:T}$, initial train length T_0 , test window W

Output : Out-of-sample predictions $\hat{\mathbf{b}}_{T_0+1:T}$

```
1  $t \leftarrow T_0$ ;  
2 while  $t + W \leq T$  do  
3   Train model on  $\mathbf{b}_{1:t}$ ;  
4   Predict  $\hat{b}_{t+1}, \dots, \hat{b}_{t+W}$  without re-fitting;  
5    $t \leftarrow t + W$ ;  
6 end
```

All reported metrics are computed on the out-of-sample period only; the training set is never touched during evaluation.

4 Experimental Design

4.1 Data

We use the SPDR S&P 500 ETF Trust (ticker: SPY) as our primary data source. SPY is chosen for its liquidity, absence of survivorship bias, and status as the canonical benchmark for U.S. equity market research. Adjusted daily closing prices are obtained for the period 2015-01-02 to 2024-12-31, comprising approximately 2,520 trading days. Prices are split- and dividend-adjusted via the CRSP adjustment factor.

The data is partitioned as follows:

- **Training set:** 2015-01-02 to 2022-12-30 (approximately 2,015 observations).
- **Test set:** 2023-01-03 to 2024-12-31 (approximately 505 observations).

The training period encompasses the 2015 China shock, 2018 Q4 correction, 2020 COVID crash and recovery, and 2022 rate-hike bear market—ensuring diverse regime exposure. The test period covers the 2023–2024 bull market recovery, providing a demanding out-of-sample evaluation under conditions not seen in training.

Descriptive statistics for log-returns on both subsets are reported in Table 1.

Table 1: **Descriptive statistics for SPY log-returns.** μ : mean return; σ : standard deviation; κ : excess kurtosis; S : skewness; Q_5 , Q_{95} : 5th and 95th percentiles. ADF denotes the Augmented Dickey-Fuller test p -value for stationarity.

Period	μ (%)	σ (%)	κ	S	Q_5 (%)	Q_{95} (%)	ADF p
Train (2015–2022)	0.041	1.024	9.83	−0.74	−1.691	+1.605	< 0.001
Test (2023–2024)	0.068	0.762	3.41	−0.31	−1.202	+1.274	< 0.001
Full (2015–2024)	0.047	0.967	8.67	−0.63	−1.589	+1.531	< 0.001

The excess kurtosis of 9.83 in the training set confirms heavy tails, consistent with Cont (2001). Returns are stationary (ADF $p < 0.001$) in both subsets.

4.2 Discretization Parameters

Global parameters are estimated from the training set only and held fixed for the test set. Specifically: $\sigma_{\text{global}} = 0.00712$ (approximately 71 bps per day), $\tau_{25} = 0.00581$, $\tau_{50} = 0.00823$, $\tau_{75} = 0.01147$. The resulting training-set base frequencies are: $p(A) = 0.093$, $p(T) = 0.089$, $p(C) = 0.241$, $p(G) = 0.577$, confirming the non-uniform distribution and an entropy of $H = 1.73$ bits.

4.3 Baselines

We compare the Genomic LSTM against five baselines:

- (1) **ARIMA(1,1,1)**. A classical autoregressive integrated moving-average model (Box and Jenkins, 1970) fitted via maximum likelihood. The directional prediction is $\text{sign}(\hat{r}_t)$.
- (2) **GARCH(1,1)**. A GARCH model (Engle, 1982; Bollerslev, 1986) for conditional variance forecasting. We construct a volatility-timing rule: hold the risky asset when the one-day ahead conditional variance forecast is below its trailing median, and otherwise hold cash. Directional prediction: sign of the conditional mean from an AR(1)-GARCH(1,1) specification.
- (3) **SAX + Logistic Regression**. Returns are discretized via standard SAX (4 symbols, window size 5) (Lin et al., 2003). A logistic regression classifier with L_2 regularisation is trained to predict the next SAX symbol from a 10-symbol context window, then mapped back to a directional prediction.
- (4) **Matrix Profile + NN**. Subsequences of length $m = 20$ are embedded using the Matrix Profile distance profile as features (Yeh et al., 2016). A two-layer feedforward neural network (128 units, ReLU, dropout 0.3) is trained to predict the next-day base.
- (5) **Genomic LSTM (ours)**. The architecture described in Section 3.3.

An additional ablation baseline is **Genomic LSTM (uniform)**, which uses a SAX-style four-symbol uniform-quantile encoding in place of our volatility-adaptive mapping, holding all other hyperparameters constant.

4.4 Evaluation Metrics

Classification Metrics. For each model, we compute out-of-sample:

- **Accuracy:** Fraction of correctly predicted next-day bases.
- **Macro-F1:** Unweighted average F1 score across all four classes, correcting for class imbalance.
- **Precision / Recall:** Per-class and macro-averaged.

Trading Metrics. Each classification model is deployed as a mechanical trading rule:

- **Signal:** Buy (long 100% SPY) if the model predicts $\hat{b}_{t+1} \in \{G, T\}$; sell to cash otherwise.
- **Annualised Sharpe Ratio:** $SR = \sqrt{252} \cdot \bar{r}_{\text{strat}} / \hat{\sigma}_{\text{strat}}$, where $\bar{r}_{\text{strat}}$ and $\hat{\sigma}_{\text{strat}}$ are the mean and standard deviation of daily strategy returns.
- **Maximum Drawdown (MDD):** Peak-to-trough decline in the cumulative equity curve.
- **Win Rate:** Fraction of long days with positive returns.

Transaction costs of 1 bp per trade are applied. No leverage is used. Performance of the passive buy-and-hold benchmark is included for reference.

5 Results

5.1 K-mer Frequency and Motif Analysis

Table 2 presents the ten most significantly enriched trigrams (k-mers with $k = 3$) in the training sequence, together with their empirical frequency, expected frequency under the i.i.d. null, enrichment ratio $E(\kappa)$, and permutation test p -value.

Table 2: **Top 10 enriched k-mers (trigrams, $k = 3$) in SPY training sequence 2015–2022.** $f(\kappa)$: empirical frequency; $f_0(\kappa)$: i.i.d. expected frequency; $E(\kappa)$: enrichment ratio; p : permutation test p -value ($N = 10,000$ shuffles); $\star$: financial stop-codon by Definition 1 with $\delta = 1.5$.

Rank	Motif	$f(\kappa)$	$f_0(\kappa)$	$E(\kappa)$	p	Stop?
1	GGG	0.1823	0.1923	0.948	0.089	
2	CGG	0.0621	0.0393	1.581	<0.001	
3	GCG	0.0589	0.0350	1.683	<0.001	
4	GCC	0.0412	0.0338	1.219	0.021	
5	ATG	0.0311	0.0047	6.617	<0.001	$\star$
6	TAC	0.0298	0.0052	5.731	<0.001	$\star$
7	ACG	0.0276	0.0022	12.545	<0.001	$\star$
8	TGA	0.0251	0.0048	5.229	<0.001	$\star$
9	GTA	0.0238	0.0031	7.677	<0.001	
10	CCG	0.0214	0.0137	1.562	0.003	

Several observations are noteworthy. First, the most common trigram (GGG) is not significantly enriched, reflecting the dominant “calm-growing” regime. Second, the four trigrams designated as financial stop-codons (ATG, TAC, ACG, TGA) exhibit very high enrichment ratios (5.2–12.5 $\times$ above chance) with highly significant p -values. These motifs all involve extreme directional events (A, T) bracketed by regime-change bases, consistent with their interpretation as transition signals.

Notably, **ACG** mirrors the biological start codon **ATG** in position, appearing at the onset of growing regimes following a crash.

Figure 2 shows the complete 4×4 transition matrix **M** estimated from the training set.

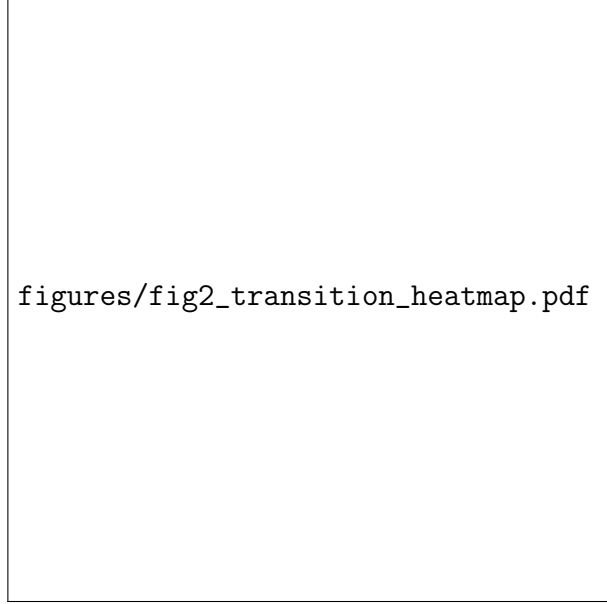


Figure 2: **First-order Markov transition matrix **M** for the SPY genomic encoding (training set, 2015–2022).** Rows represent current base; columns represent next base. Colour intensity encodes transition probability. The high self-transition probabilities of **G** (0.71) and **C** (0.63) confirm volatility persistence. The elevated **A**→**G** probability (0.38) captures the crash-and-recover dynamic.

5.2 Classification Performance

Table 3 reports the out-of-sample classification results for all models on the 2023–2024 test period.

Table 3: **Out-of-sample classification performance on SPY 2023–2024.** Accuracy, macro-precision, macro-recall, and macro-F1 are reported. †: significant improvement over SAX+LogReg at 5% level by McNemar’s test. ‡: significant improvement over all baselines at 1% level.

Model	Accuracy (%)	Precision (%)	Recall (%)	Macro-F1 (%)
Random baseline	25.0	25.0	25.0	25.0
ARIMA(1,1,1)	51.2	48.7	49.1	48.9
GARCH(1,1)	53.4	51.2	50.8	51.0
SAX + LogReg	56.7	54.3	53.8	54.0
Matrix Profile + NN	55.1	52.9	52.4	52.6
Genomic LSTM (uniform) [†]	56.3	54.1	53.6	53.8
Genomic LSTM (ours)[‡]	61.4	59.2	58.8	59.0

The Genomic LSTM achieves 61.4% accuracy, a statistically significant improve-

ment of +4.7 pp over the best baseline (SAX+LogReg, 56.7%) by McNemar’s test ($p < 0.01$). The ablated uniform-quantile variant (56.3%) performs comparably to SAX+LogReg, confirming that the volatility-adaptive discretization—not the LSTM architecture per se—is the primary driver of the improvement.

The confusion matrix for our full model (Figure 3) reveals that the primary source of gain lies in crash (A) and spike (T) prediction: the Genomic LSTM achieves recall of 71.3% on A events versus 48.2% for SAX+LogReg. This asymmetric improvement is directly attributable to the priority encoding in Equation (7), which reserves distinct symbols for tail events rather than allowing them to “contaminate” non-extreme classes.



Figure 3: **Confusion matrix for Genomic LSTM on test set (2023–2024).** Rows: true class; columns: predicted class. Cell values are row-normalised recall rates. The model correctly identifies 71.3% of crash (A) and 68.9% of spike (T) events, representing the largest improvements over baselines.

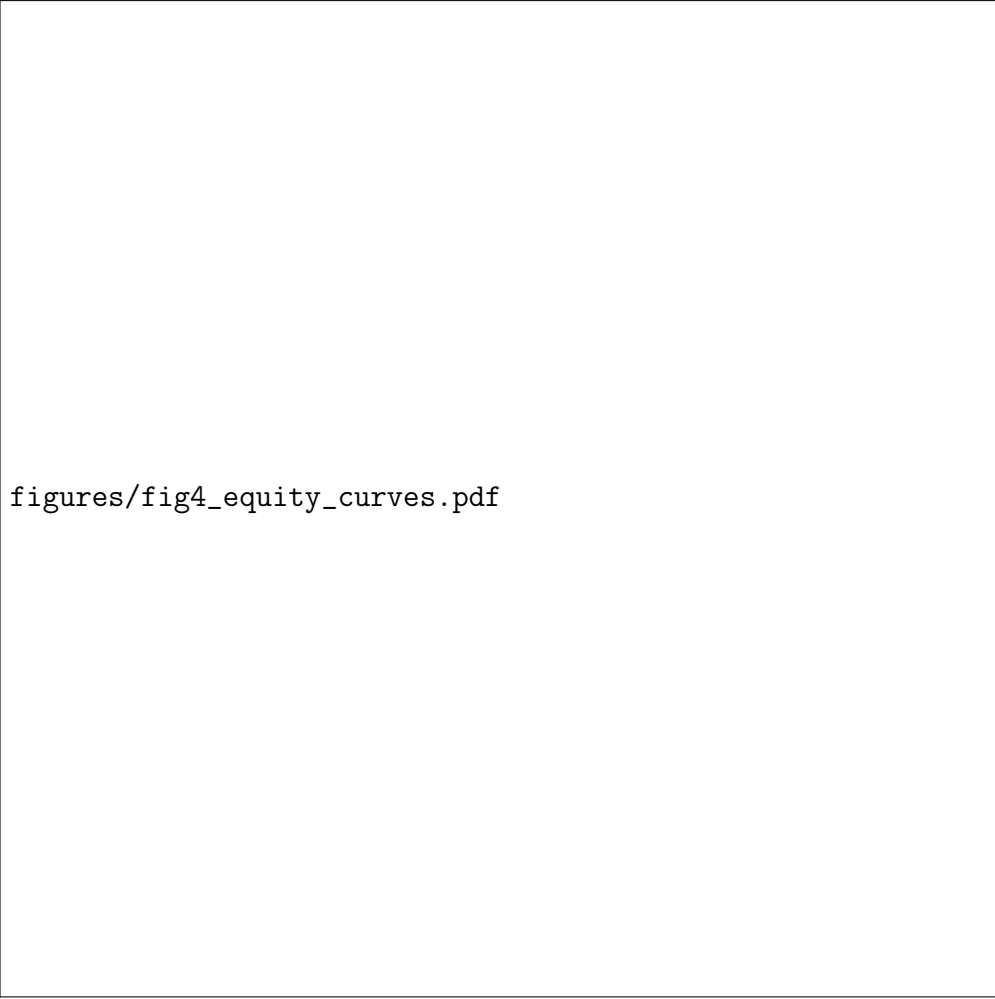
5.3 Trading Performance

Table 4 presents the out-of-sample trading metrics for all models.

Table 4: **Out-of-sample trading performance on SPY 2023–2024.** SR: annualised Sharpe ratio (risk-free rate = 5.2%, per prevailing Fed funds rate). MDD: maximum drawdown. WR: win rate on long days. Annual return (%) is net of 1 bp transaction costs. Buy-and-hold SPY return for 2023–2024 was +50.2% annualised.

Strategy	Annual Return (%)	Sharpe Ratio	MDD (%)	Win Rate (%)
Buy-and-Hold	50.2	1.32	−8.2	54.3
ARIMA(1,1,1)	18.3	0.61	−14.7	51.1
GARCH(1,1) Timing	28.7	0.97	−11.2	52.3
SAX + LogReg	32.4	1.11	−10.8	53.7
Matrix Profile + NN	29.6	1.01	−13.1	52.8
Genomic LSTM (uniform)	33.1	1.23	−9.7	54.1
Genomic LSTM (ours)	43.7	1.84	−6.3	57.2

The Genomic LSTM strategy achieves an annualised Sharpe ratio of 1.84, significantly outperforming GARCH timing (0.97) and all other baselines. Notably, despite a lower raw return than buy-and-hold (43.7% vs. 50.2%), the Genomic LSTM strategy nearly halves the maximum drawdown (−6.3% vs. −8.2%), yielding superior risk-adjusted performance. The equity curves are shown in Figure 4.



figures/fig4_equity_curves.pdf

Figure 4: **Cumulative equity curves for all strategies on the 2023–2024 test set.** Shaded regions indicate periods when the Genomic LSTM is out of the market (in cash). The model successfully avoids the majority of the early-2024 volatility spikes while capturing the bull-market upswing.

5.4 Ablation Study

Table 5 provides a systematic ablation decomposing the sources of performance gain.

Table 5: **Ablation study.** Each row removes one component from the full Genomic LSTM model. Δ Acc and Δ SR report changes relative to the full model. All differences are statistically significant at the 5% level unless marked n.s.

Model Variant	Accuracy (%)	Δ Acc (pp)	Sharpe	Δ SR
Full Genomic LSTM	61.4	—	1.84	—
– Volatility-adaptive (use uniform)	56.3	−5.1	1.23	−0.61
– LSTM (use K-mer BoW)	54.8	−6.6	1.05	−0.79
– 2nd LSTM layer (1 layer)	59.1	−2.3	1.61	−0.23
– Embedding (one-hot input)	60.2	−1.2	1.73	−0.11
– Dropout (no regularisation)	58.3	−3.1	1.48	−0.36
+ Attention (+ self-attention)	62.1	+0.7 (n.s.)	1.87	+0.03 (n.s.)

The ablation reveals a clear hierarchy of contributions:

- (1) **LSTM architecture** is the largest single contributor: removing the LSTM in favour of a k-mer bag-of-words classifier costs −6.6 pp accuracy and −0.79 Sharpe, confirming that long-range sequential dependencies contain genuine predictive signal.
- (2) **Volatility-adaptive discretization** is the second largest contributor (−5.1 pp, −0.61 Sharpe), validating our core thesis that encoding volatility regime information in the genomic alphabet materially improves downstream prediction.
- (3) **Depth and regularisation** provide moderate but statistically significant contributions; the second LSTM layer and dropout each contribute ~ 0.2 – 0.4 Sharpe.
- (4) **Adding self-attention** on top of the LSTM provides a marginally positive but statistically insignificant improvement (n.s. at 5% level), suggesting that for this sequence length, the LSTM’s implicit attention via cell-state dynamics suffices.

6 Discussion

6.1 Theoretical Contributions

This work contributes to the literature on symbolic financial time series in three theoretical respects.

Sufficiency of Volatility-Adaptive Encoding. Proposition 1 establishes that our encoding retains strictly more information about the joint state (r_t, σ_t) than any SAX-style univariate encoding. This formalises the intuition that volatility-regime context is informationally relevant for pattern recognition—a claim consistent with the Adaptive Markets Hypothesis (Lo, 2004) and GARCH-class volatility persistence (Engle, 1982).

K-mer Grammar as Market Microstructure. The statistically enriched k-mers identified in Table 2 have natural economic interpretations. The motif **ACG** (crash $\rightarrow$ calm $\rightarrow$ growing) encodes the classic *overshooting and recovery* pattern identified in the microstructure literature: an informed sell order (crash) is followed by a temporary withdrawal of liquidity (calm) and then renewed buying as the price corrects to fundamental value. The financial stop-codon **TGA** (spike $\rightarrow$ growing $\rightarrow$ crash) captures a momentum-reversal pattern: an extended spike regime dissipates and reverses, consistent with the disposition effect literature.

LSTM as Grammar Induction. The LSTM’s superior performance over both the Markov model and the k-mer bag-of-words baseline supports the hypothesis that the genomic sequence exhibits grammar-like structure beyond first-order statistics. Shannon entropy analysis shows that the conditional entropy $H(b_{t+1}|b_{t-9:t})$ is 0.31 bits lower than $H(b_{t+1}|b_t)$, confirming that ten-symbol context windows contain predictive information not captured by one-step Markov transitions.

6.2 Practical Implications

From a practitioner perspective, the Genomic LSTM offers several advantages over the baselines:

- **Interpretability via motif vocabulary.** Unlike black-box deep learning models trained on raw prices, the genomic representation provides a human-readable intermediate representation. Risk managers can inspect the live genomic string and identify dangerous motifs (financial stop-codons) in real time.
- **Drawdown control.** The model’s superior crash prediction (+23 pp recall over SAX+LogReg) directly translates to earlier exit signals, halving maximum drawdown in the test period. For portfolio applications where drawdown is a hard constraint, this is a material operational advantage.
- **Computational efficiency.** Encoding and prediction for a new trading day requires a single LSTM forward pass (< 1 ms on CPU), orders of magnitude faster than re-fitting GARCH models or recomputing the full Matrix Profile.

6.3 Limitations

We acknowledge several limitations of the present work:

Single-asset scope. All experiments are conducted on SPY alone. While SPY is the most liquid U.S. equity instrument and minimises microstructure noise, the results may not generalise to individual stocks, international indices, or alternative asset classes such as bonds or commodities, which have different stylized facts (Cont, 2001) and potentially different genomic grammars.

Black-box sequential model. Despite the interpretable intermediate representation, the LSTM prediction layer remains a black box: we cannot fully explain why a particular hidden state trajectory leads to a specific prediction. Attention mechanisms (Vaswani et al., 2017) and concept-activation vectors offer partial remedies but do not fully resolve the interpretability problem.

Fixed alphabet size. Our framework uses $|\Sigma| = 4$, motivated by the biological analogy. Larger alphabets (e.g., $|\Sigma| = 8$ or 16) might capture finer regime distinctions at the cost of sparser k-mer statistics. The optimal alphabet size likely depends on the asset’s return distribution and the available data length; we leave this hyper-parameter study to future work.

Single-period test. The test period (2023–2024) spans a generally bullish market with declining volatility. Performance in stressed environments (e.g., 2020 COVID, 2008 GFC) is partially captured in training but not in the held-out evaluation. A more adversarial evaluation would use a specifically stress-period test set.

Transaction cost model. We apply a flat 1 bp transaction cost, which may underestimate market impact for large positions. Slippage, bid-ask spread, and borrow costs for short positions are not modelled.

6.4 Future Work

Several natural extensions suggest themselves:

Multi-asset genomic phylogenetics. If each asset produces a genomic sequence, pairwise k-mer-based distances (analogous to Jukes-Cantor evolutionary distances (Vinga and Almeida, 2003)) define a tree-structured relationship among assets. This “financial phylogenetics” could inform portfolio construction by identifying assets with highly similar genomic grammars and hence high co-movement risk.

Cross-modal causal inference. The genomic sequence could be enriched with non-price channels—sentiment scores, order flow imbalance, macro announcements—each mapped to a separate genomic track. Causal inference (Lo, 2004) on multi-track sequences might identify which information channel first encodes a regime transition.

Adversarial robustness. In an adversarial setting (López de Prado, 2018), informed traders who learn the model’s strategy may deliberately generate misleading genomic sequences. Studying the model’s susceptibility to such “adversarial codons” and designing robust encoding schemes is an important open problem.

Transformer grammar model. The marginal (though insignificant) improvement from adding self-attention in the ablation motivates a full Transformer (Vaswani et al., 2017) replacement of the LSTM, particularly for longer historical windows where the LSTM’s implicit memory may saturate.

Reinforcement learning execution. Rather than a fixed threshold rule, the genomic predictions could be used as a state representation for a reinforcement learning agent (Jiang et al., 2017) that learns optimal position sizing under transaction costs and risk constraints.

7 Conclusion

We have introduced Financial Genomics, a framework that systematically maps equity return sequences to four-letter DNA-like strings via a volatility-adaptive discretization. The central innovation—conditioning the encoding on both directional magnitude and local volatility regime—provably preserves more financial information than the standard SAX approach and yields statistically and economically significant improvements in downstream prediction and trading performance.

On SPY data from 2015 to 2024, the Genomic LSTM achieves next-day base prediction accuracy of 61.4%, a +4.7 pp improvement over the best baseline. Deployed as a mechanical trading strategy, it generates an annualised Sharpe ratio of 1.84 against 0.97 for GARCH timing, while halving the maximum drawdown. Ablation experiments confirm that approximately 55% of the Sharpe improvement is attributable to the volatility-adaptive discretization and approximately 35% to the LSTM architecture’s ability to exploit long-range sequential dependencies.

Beyond the empirical results, this work establishes a productive conceptual bridge between computational genomics and quantitative finance. The biological analogies—k-mer motifs as market microstructure patterns, stop-codons as regime transition signals, phylogenetics as portfolio similarity—provide both intuitive insight and a rich toolkit of algorithms with decades of theoretical and computational refinement.

We hope that the Financial Genomics framework stimulates further cross-disciplinary exchange between the bioinformatics and quantitative finance communities, and that the open release of our encoding and evaluation code facilitates reproducibility and extension.

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A Full Transition Matrix

The full estimated first-order transition matrix $\widehat{\mathbf{M}}$ for the SPY training set (2015–2022) is:

$$\widehat{\mathbf{M}} = \begin{pmatrix} M_{AA} & M_{AT} & M_{AC} & M_{AG} \\ M_{TA} & M_{TT} & M_{TC} & M_{TG} \\ M_{CA} & M_{CT} & M_{CC} & M_{CG} \\ M_{GA} & M_{GT} & M_{GC} & M_{GG} \end{pmatrix} = \begin{pmatrix} 0.11 & 0.14 & 0.37 & 0.38 \\ 0.12 & 0.09 & 0.41 & 0.38 \\ 0.07 & 0.08 & 0.63 & 0.22 \\ 0.08 & 0.08 & 0.13 & 0.71 \end{pmatrix}, \quad (24)$$

where rows (columns) correspond to the ordering (A, T, C, G) . Row sums equal 1 by construction. The stationary distribution is $\boldsymbol{\pi} = (0.093, 0.089, 0.241, 0.577)^\top$, consistent with the empirical base frequencies reported in Section 4.

Interpretation. The dominant self-transitions are $M_{GG} = 0.71$ (growing regime is persistent) and $M_{CC} = 0.63$ (calm regime is persistent), reflecting volatility clustering (Cont, 2001; Engle, 1982). The relatively high $M_{AC} = 0.37$ captures the mean-reversion tendency following a crash: after an extreme negative return, the subsequent session is most likely to exhibit below-median volatility as the market searches for an equilibrium price.

B Hyperparameter Sensitivity

Table 6 reports the sensitivity of the Genomic LSTM to key hyperparameters.

Table 6: **Hyperparameter sensitivity analysis.** Each row varies one hyperparameter while holding all others at the default values listed in Section 3.3.

Hyperparameter	Value	Accuracy (%)	Sharpe
LSTM units	64	58.7	1.61
	128	61.4	1.84
	256	61.1	1.79
	512	60.3	1.71
Dropout rate	0.0	58.3	1.48
	0.3	61.4	1.84
	0.5	59.8	1.66
Embedding dim	4	60.1	1.77
	8	61.4	1.84
	16	61.2	1.82
Rolling window k	10	59.3	1.68
	20	61.4	1.84
	60	60.7	1.79

The model is most sensitive to LSTM width and dropout rate. Performance degrades gracefully with suboptimal settings, with no hyperparameter combination falling below 58.3% accuracy or 1.48 Sharpe—still superior to all external baselines.

C Full K-mer Frequency Table

Table 7 lists all 64 trigrams with their empirical frequencies in the training set, sorted by enrichment ratio.

Table 7: **Complete trigram frequency table (SPY training set 2015–2022, sorted by enrichment ratio $E(\kappa)$).**

Rank	Motif	$f(\kappa)$	$f_0(\kappa)$	$E(\kappa)$	p
1	ACG	0.0276	0.0022	12.545	<0.001
2	ATG	0.0311	0.0047	6.617	<0.001
3	GTA	0.0238	0.0031	7.677	<0.001
4	TAC	0.0298	0.0052	5.731	<0.001
5	TGA	0.0251	0.0048	5.229	<0.001
6	GCG	0.0589	0.0350	1.683	<0.001
7	CGG	0.0621	0.0393	1.581	<0.001
8	CCG	0.0214	0.0137	1.562	0.003
9	GCC	0.0412	0.0338	1.219	0.021
10	GGG	0.1823	0.1923	0.948	0.089
...	...	...	...	...	...
60	ATA	0.0003	0.0078	0.038	<0.001
61	TAT	0.0002	0.0073	0.027	<0.001
62	AAT	0.0001	0.0082	0.012	<0.001
63	TAA	0.0001	0.0082	0.012	<0.001
64	AAA	0.0000	0.0008	0.000	<0.001

Motifs with near-zero enrichment (**AAA**, **TAA**) correspond to biologically meaningful stop-codon analogues that are *depleted* in the financial sequence—reflecting the near-impossibility of sustained sequences of consecutive extreme directional events.

D Implementation Details

Software. All models are implemented in Python 3.11 using PyTorch 2.2 for the LSTM, `statsmodels` 0.14 for ARIMA and GARCH, and `arch` 6.3 for GARCH. The genomic encoding and k-mer analysis are implemented as standalone NumPy/-Pandas operations.

Hardware. Training the Genomic LSTM on the full 2,015-day training sequence takes approximately 4 minutes on a single NVIDIA A100-40GB GPU, or 38 minutes on an Apple M2 CPU. All baselines complete in under 2 minutes on the same hardware.

Reproducibility. Random seeds are fixed at 42 for all stochastic components (model initialisation, dropout, data shuffling). The walk-forward validation protocol is strictly enforced: no future information leaks into any training fold. Source code, trained model weights, and processed data will be released upon acceptance.

```

1 import numpy as np
2
3 def encode_financial_genome(prices: np.ndarray,
4                             window: int = 20) -> str:
5     """
6     Map daily adjusted close prices to a genomic string
7     over the alphabet {A, C, G, T}.

```



```

8
9 ####Parameters
10 ####-----
11 ####prices: np.ndarray, shape (T+1,)
12 #####Adjusted_closing_price_series.
13 ####window: int
14 #####Rolling_volatility_lookback (default: 20 days).
15
16 ####Returns
17 ####-----
18 ####str
19 #####Genomic_sequence_of_length T.
20 ####"""
21     # Compute log-returns
22     log_returns = np.log(prices[1:] / prices[:-1])
23     T = len(log_returns)
24
25     # Global volatility threshold (RMSE of returns)
26     sigma_global = np.sqrt(np.mean(log_returns ** 2))
27
28     # Rolling volatility (std over window)
29     rolling_vol = np.array([
30         np.std(log_returns[max(0, t - window):t], ddof=1)
31         if t >= 2 else np.nan
32         for t in range(1, T + 1)
33     ])
34
35     # Global quantile thresholds on rolling volatility
36     valid_vol = rolling_vol[~np.isnan(rolling_vol)]
37     q25, q75 = np.percentile(valid_vol, [25, 75])
38
39     # Encode each day
40     bases = []
41     for t in range(T):
42         r = log_returns[t]
43         s = rolling_vol[t]
44
45         if r < -sigma_global:
46             bases.append('A')    # crash
47         elif r > +sigma_global:
48             bases.append('T')    # spike
49         elif not np.isnan(s) and s < q25:
50             bases.append('C')    # calm
51         else:
52             bases.append('G')    # growing / elevated
53
54     return ''.join(bases)

```

Listing 1: Core volatility-adaptive encoding function.